So bothering, everyone wants to take a course with me

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1 Description

During the summer vacation, A large number of people become your nice buddy and you are namely **the core of the department** right now. Everyone wants to have the same course with you. But everyone is so annoying and they have different requirements. For example, Frankie states that I want a course with easy homework, If it does not, then at least the teacher must be handsome. Amber states that I want a course which is inspiring, If it does not, then at least it must locates at EE building NO.2. The question is: how can we find a course that satisfies all of the people? Does such a course exist?

2 Input

Input Format

We assume the course has N attributes represented as a boolean array (e.g., whether the homework is easy or not, whether the teacher is handsome or not, etc.). Each attribute x_i $(1 \leq i \leq N)$ can be either true or false.

The input consists of M requirements, each expressed as a disjunction of two literals.

```
N M
a_1 b_1
a_2 b_2
⋮
a_M b_M
```

where:

- N — the number of attributes.
- M — the number of requirements (clauses).
- Each pair (a_i, b_i) represents one requirement $(\ell_{i1} \vee \ell_{i2})$.
- If $a_i > 0$, it means literal x_{a_i}; if $a_i < 0$, it means $\neg x_{|a_i|}$.
- The same rule applies for b_i.

In other words, the overall condition can be written as:

$$(\ell_{11} \vee \ell_{12}) \wedge (\ell_{21} \vee \ell_{22}) \wedge \cdots \wedge (\ell_{M1} \vee \ell_{M2})$$

and you need to determine whether there exists a truth assignment for all x_i that satisfies every clause.

3 solution

The problem is actually the well-known 2-SAT problem. To begin with

1. For a clause $(a \vee b)$, it is equivalent to $(\neg a \to b) \wedge (\neg b \to a)$.
2. Therefore, we can build a directed graph G with $2N$ nodes, each variable x_i has two nodes: $2i$ for positive and $2i + 1$ for negative.
3. For each clause $(a \vee b)$, we add two directed edges $\neg a \to b$ and $\neg b \to a$ to the graph.
4. After the cnstruction of the graph, we compute the SCCs of the graph.
5. If for any variable x_i, both nodes $2i$ and $2i+1$ belong to the same SCC, then the 2-SAT instance is unsatisfiable. (since it implies that x_i must be both true and false).

4 pseudo code

```
Input: N (number of variables), M (number of clauses), each clause
       (a_i, b_i)
Output: Whether there exists a satisfying assignment
Build a directed graph G with 2N nodes;
for i ← 1 to M do
    read a, b;
    u ← node(a);
    v ← node(b);
    add edge ¬u → v to G;
    add edge ¬v → u to G;
end
Compute strongly connected components (SCC) of G;
for i ← 0 to N − 1 do
    if SCC(2i) == SCC(2i + 1) then
        return NO;
    end
end
return YES;
```

Algorithm 1: 2-SAT satisfiability checking

4.1 time complexity

To build the graph G, we need to process M clauses, each adding two edges, resulting in $O(M)$ time.
Computing SCC requires $O(N + M)$ time since we have $2N$ nodes and $2M$ edges. and DFS requires $O(N + M)$ time as well.
Finally, checking each variable for conflicts takes $O(N)$ time.
Overall, the time complexity of the algorithm is $O(N + M)$.

選課時間

Figure 1: something for fun